Supplementary material

For "When Is a Call–Put Implied-Volatility Spread Measurable? An Abstention-Aware Protocol Validated Against a Known Truth". These appendices are not typeset with the article; they are published online alongside it.

Table S1. Certified coverage by strata, development draw

Family	Bucket	n	Certified coverage (%)	Certified MAE (pp)	Raw-midpoint MAE (pp)	Raw MAE on abstained (pp)
Moneyness	K/S=0.90	179	0.0	—	0.054	0.054
Moneyness	K/S=0.95	179	13.4	0.010	0.028	0.030
Moneyness	K/S=1.00	179	13.4	0.009	0.024	0.027
Moneyness	K/S=1.05	178	13.5	0.007	0.041	0.047
Moneyness	K/S=1.10	179	0.0	—	2.508	2.508
Maturity	14d	298	0.0	—	1.504	1.504
Maturity	30d	298	0.0	—	0.068	0.068
Maturity	60d	298	24.2	0.009	0.019	0.022
Half-spread	1 tick half-spread	299	24.1	0.009	0.020	0.023
Half-spread	2 tick half-spread	348	0.0	—	0.062	0.062
Half-spread	3 tick half-spread	99	0.0	—	3.568	3.568
Half-spread	wide session	148	0.0	—	0.644	0.644
True magnitude	0.1-1 pp	299	16.1	0.010	0.022	0.024
True magnitude	<0.1 pp	238	0.0	—	0.052	0.052
True magnitude	>=1 pp	357	6.7	0.007	1.275	1.366

Note: n is the number of attempted pairs in the bucket. Coverage is the share of the bucket the protocol certifies at 0.10 percentage points. MAE is the mean absolute error against the benchmark's defined truth, in percentage points of volatility, computed on the pairs each method can use; a dash means the method certified nothing in that bucket. The rightmost column is the raw midpoint's error on the pairs the protocol declined to certify.

Table S2. The declared settings

Setting	Value
Inversion map	European Black–Scholes–Merton, common to both legs
Root bracket	decimal IV in $[10^{-8}, 5]$, bisection, ≤ 64 iterations, no clipping
Near-bound guard	prices within 10^{-10} of a European bound are rejected, not inverted
Resolution scale	1 tick of out-of-the-money-equivalent time value
Synchronization tolerance	2 s for call–put, call–spot and put–spot lags
Tolerance grid	0.05 / 0.10 / 0.25 percentage points, all reported
Primary tolerance	0.10 percentage points, declared in the design addendum

Setting	Value
Sign band	0.05 percentage points
Perturbation set	endpoint widened by half a tick; spot +1 bp; rate +5 bp; spot timestamp −3 s; resolution scale halved to 0.5 tick; synchronization tolerance relaxed to 5 s
Period split	primary = days 0–14, holdout = days 15–29
Inference unit	underlying-day block, 2 000 resamples, seed 20261022

Note: every value is declared in the design addendum. The addendum's first version in the repository history is dated after the replication sweep outputs were generated; this paper does not claim that the values were fixed before those outputs were read (Section 5.1).

Appendix A. The protocol and the instrument in one page

A.1 Stages. Contract match → timing check → quote validity → input validity → midpoint bounds → endpoint inversion → time-value resolution → finite-shock resolution → tolerance certification.

A.2 Statuses. Seven terminal statuses, mutually exclusive, summing to the attempted count. Reason codes stored separately and never substituted for a status.

A.3 Inversion. European Black–Scholes–Merton for both legs, decimal IV in $[10^{-8}, 5]$, bisection with at most 64 iterations, no clipping, near-bound guard at 10^{-10}.

A.4 Resolution. Time value of at least the resolution scale; valid roots under both declared shocks; a finite endpoint range over the four bid–ask combinations, one-sided where an endpoint has no root, never imputed.

A.5 Certification. The endpoint range of the CPIV is narrower than the declared tolerance.

A.6 Inference. Underlying-day block bootstrap, 2 000 resamples, seed 20261022. The same block device is used for the null in the selection frontier.

A.7 Generator. Two underlyings at spot 100 and 250; 30 trading days each; maturities of 14, 30 and 60 days; strikes at 90 %, 95 %, 100 %, 105 % and 110 % of spot; a continuously compounded rate of 5 % and no dividend yield; base volatility of 22 % with a term slope of +0.12 percentage points per day and a log-moneyness skew of −1.10; a tick of 0.01, a half-spread of one tick, one extra tick in the wings and at the short end, doubled on every seventh session; timestamp jitter of ±0.35 s; and injected defects at declared rates — stale quotes 1.5 %, crossed quotes 0.8 %, out-of-bounds quotes 0.5 %, contract mismatches 0.5 % and missing legs 0.4 %. The put leg is priced on an 800-step binomial lattice, so the truth carries a genuine early-exercise premium.

Appendix B. The replication design in one page

B.1 Seeds. 23 replication seeds, 20261022 + 7919k for k = 1 … 23, plus the development draw at k = 0 which is excluded from every tally.

B.2 Specifications. 17, one factor at a time from the base market: nine market specifications, seven protocol specifications and the base itself.

B.3 Frontier. 2 000 Monte Carlo trials at each of seven evaluation points — 0.05, 0.075, 0.10, 0.15, 0.20, 0.25 and 0.50 percentage points — against a pair null (uniform without replacement) and a day-block null (whole underlying-days with replacement, partial final

block taken at random). Add-one p-values. That axis contains the protocol tolerance grid of Table S2 plus intermediate points so the curve can be drawn; it is an evaluation axis, not the grid. The grid itself is 0.05, 0.10 and 0.25 percentage points, and the 0.50-point evaluation is an additional tolerance specification used in the sweep (Section 5.3), not part of the grid. Certification and the reported coverages are governed by the grid alone, and the decision rules of the design addendum read the frontier at the primary tolerance only, 0.10 pp. Table 3 shows the grid points that certify plus two intermediate points; the other rows are in the released artifact.

B.4 Tally. `verdict.py` encodes the six rules of the design addendum and reads only the replication-seed records of the sweep outputs. R3 and R6 are aggregated over the 23 seeds before they are tested: R3 uses each market specification's mean certified coverage, and R6 uses each perturbation's mean change in certified coverage. The per-draw reading gives the same outcome for both. The development draw is excluded from the tally.

B.5 Artifacts. The run manifests record the configuration, the seed, the environment and the output hashes of every run reported in the article. The per-draw sweep outputs, the specification grid and the rule tally are released as aggregate CSV and JSON files; no market data is included because none was used.